

Colnar, Audrey M., and Wayne G. Landis. "Conceptual model development for invasive species and a regional risk assessment case study: The European green crab, *Carcinus maenas*, at Cherry Point, Washington, USA." *Human and Ecological Risk Assessment* 13, no. 1 (2007): 120-155.

- The threat of European Green Crabs was assessed using an adaptation Relative Risk Model
- Eel grass habitats and Dungeness crab populations were evaluated using 2 postulated scenario's: then current conditions in 2004 and a future El Nino season
- The end results showed that an El Nino situation creates a much more negatively impactful result on the environment around Washington.
 - This can be used to determine how the impact of green crabs is both detrimental to the PNW and can be exacerbated by environmental factors not usually associated with mortality in normal environments
 - This works with any paper that seeks to evaluate the need for or preparations for a structured removal of green crabs in a certain location, in order to relieve stress of the shore crabs

Ray, Gary L. "Invasive marine and estuarine animals of the Pacific northwest and Alaska." (2005).

- Invasive species get into the PNW by the hundreds every year but a few have the potential to be detrimental to the environment
- Not just the environment but the US Army Corp of Engineers can also be drastically affected
- Previous works have shown how the environmental destruction caused by invasive species has hindered work by the USACE in the past
 - This could be a very useful study for 2 reasons: one is that it helps clear any public misconception that invasive species are these devastating situations that are destined to destroy the environment. Two is that it's not just the environment that gets affected, its daily human life that gets affected, the military, and as a consequence, the general public as well.
 - This paper works with previous work and evidence to bring to light a specific group of the PNW that is affected by invasive species. While it doesn't specify European green crabs as a direct source of problems, this would help any paper talking about invasive species in any capacity in the discussion section when talking about consequences that may arise from neglect

Ens, Nicholas J., Emily G. Lim, Brett R. Howard, and Tao M. Eastham. "A comparison of the predatory impacts of an invasive and native crab species using a functional response approach." *Biological Invasions* 23, no. 7 (2021): 2329-2336.

- Ecological impacts are still not fully understood by the general public
- Compare to native rock crabs' impact on the environment to determine how bad the invasive species is to the PNW
- Oysters were fed to both crabs to determine feeding habits
- Rock crabs feed when there is high abundance and slow down when "populations" of oysters decrease. Green crabs feed even when the populations are low creating a more destabilizing impact on the environment for both the oysters and the rock crabs
- Functional response analysis is used to compare invasive with native species to determine the consequences of invasion.
 - This can be used to show how exactly invasive species harm the environment around them instead of just simply saying that they do. This could help with the public who don't fully understand the scale of the green crab's invasion
 - This also helps for any topic that talks about other species detriment due to invasive crabs
 - Results section can also use the functional response analysis to help predict future results of the green crabs and give the experiment some longevity

Walter, William L. "Green Economics: Assessing The Feasibility Of A New England Green Crab (*Carcinus Maenas*) Fishery Through Fishermen Perspectives." (2021).

- European green crab has been pervasive in New England as well as PNW
- The crab has been detrimental the aquaculture dedicated to the Maine soft shelled clam
- The warming of the Atlantic ocean is causing them to proliferate even more north up to nova scotia territory
- As a result, some fishers have conducted research on the feasibility of a green crab fishery
- Market predictions are low at the moment but recent trends suggest the price per pound of hardshell crab could make this a viable fishery
 - This would be an interesting paper because of the 2 sides it presents
 - On the one hand the pervasiveness of the green crab could harm the economy and the society built around certain fisheries but on the other hand it could be the replacement for those fisheries if stocks change in favor of the crab

- However, it also questions the need to leave an invasive species in the area all for economic benefit

Stillman, Jonathon H., and George N. Somero. "Adaptation to temperature stress and aerial exposure in congeneric species of intertidal porcelain crabs (genus *Petrolisthes*): correlation of physiology, biochemistry and morphology with vertical distribution." *Journal of experimental biology* 199, no. 8 (1996): 1845-1855.

- 2 crabs were exposed to different types of stressors
- Heartbeat, lactate accumulation and aerobic respiration of the crabs were observed when exposed to higher temperatures and tide differences
- Results found that crabs with a leg membrane have an extra respiratory structure which allows for increased adaptations to stress than other crabs
 - While these aren't the specific crabs we will work with, the testing methods and findings can help curate a plan for the stress observations of the crabs we do work with
 - Additionally, the intertidal crab stress can be correlated with similar species in shore regions as they become exposed to similar stressors and could be better managed when they become prevalent

Paganini, Adam W., Nathan A. Miller, and Jonathon H. Stillman. "Temperature and acidification variability reduce physiological performance in the intertidal zone porcelain crab *Petrolisthes cinctipes*." *Journal of Experimental Biology* 217, no. 22 (2014): 3974-3980.

- Climate data warns of increased thermal and tide stress coming to the shores of the PNW
- Porcelain crabs were exposed to current intertidal conditions and predicted conditions to observe stress response
- Under varying combinations of heat, tide level, and acidification, the crab's tolerance to heat and pH was observed and in turn, respiration rate was observed
- Results showed negative long-term effects for any ectotherms exposed to high temperature and pH
 - This could be used to apply the stress observed in these crabs to other marine shore invertebrates as their tolerances are likely similar to that of the porcelain crabs
 - The stress results could be used to determine how to manage the environment of the crabs so that the population doesn't easily get decimated

Stone, Dmitry David. "A study of the osmoregulatory role of the antennary glands in two species of intertidal crabs." PhD diss., University of British Columbia, 1962.

- 2 intertidal crabs had their osmoregulatory responses observed with their antennary glands especially observed
- In low salinities, the crabs osmoregulate heavily to keep blood hypertonic to the medium
- Their response to seasonal adaption to temperature changes
- The urine concentration especially was monitored and was found to be lower in activity in the winter than in the summer due to osmoregulation adaption to heat differences
 - The physiology of these crabs showed that winter months are a time of greater stress than summer months which could be a way to design an experiment that observes stress of the crabs
 - The data could also help management and knowledge of the shore crab populations as they're exposed to more stressors, perhaps in a relocation effort to save them from European green crabs

<https://www.seattletimes.com/seattle-news/invasive-european-green-crabs-discovered-in-was-skagit-bay/>

- Scientists have found evidence of invasive European green crabs in the PNW
- The crabs are among the most destructive invasive species in the region
- They prey on eelgrass beds and harm Dungeness crab populations which causes environmental destruction and economic destruction

<https://www.krbd.org/2025/11/12/record-number-of-invasive-green-crab-found-on-annette-island-potential-impacts-to-subsistence-fishing/>

- Invasive green crabs on the west coast cause regular economic destruction to the area
- Dungeness crab fisheries take a hit due to larval predation
- They don't seem to die off fast, which means they cause massive destruction of the environment around them